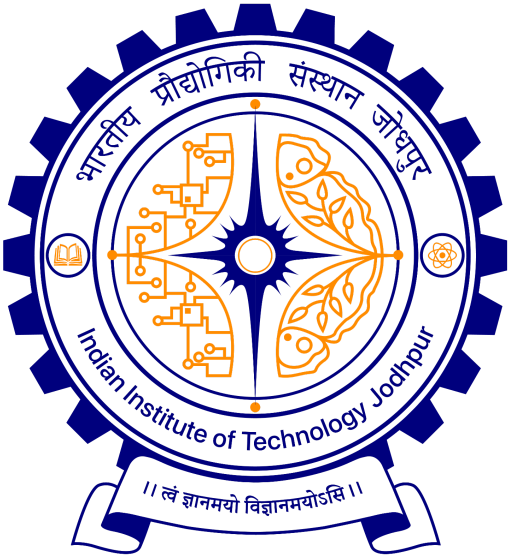
A new undergraduate program in the Department of Mathematics

B.S. in Mathematics & Computing



Department of Mathematics
Indian Institute of Technology Jodhpur 2026-27

1. Introduction and need of the programme

The B.S. in Mathematics and Computing is designed to meet the growing national and global demand for graduates who possess strong mathematical foundations and advanced computational skills. The curriculum combines foundational mathematical areas such as analysis, algebra, probability, combinatorics, and optimization with core computing subjects including algorithms, data structures, machine learning, and database systems. The programme aims to develop graduates who can approach complex problems with mathematical clarity and computational precision.

Across industry and research sectors, there is a growing demand for professionals who possess both strong mathematical foundations and advanced computational capabilities. Fields such as finance, technology, analytics, engineering, and scientific research increasingly require individuals who can design efficient algorithms, construct reliable mathematical models, analyze large-scale data, and develop scalable systems. This evolving landscape underscores the need for a programme that places mathematical thinking at the core of computational problem-solving and prepares students for roles including quant analysts, algorithm engineers, data scientists, optimization specialists, modelling experts, cryptography and security analysts.

The programme contrasts with other B.Tech. degrees in AI&DS and CSE by emphasizing the mathematical foundations and rigor necessary for developing strong analytical and computational skill sets with broad applicability. Combined with dedicated elective verticals in Data Science, Mathematical Computing, and Scientific Computing, this programme will enable students to launch into diverse career paths. Thus the programme aims to nurture graduates who are not only technically skilled but also analytical, ethical, and adaptive - capable of innovating across disciplinary boundaries and contributing meaningfully to the rapidly evolving technological landscape.

2. Objectives of the program

The B.S. in Mathematics and Computing aims to fulfill the following objectives:

- *Understanding core concepts*: Develop graduates with strong mathematical foundations and advanced computational skills, emphasizing mathematical thinking as the basis of computational problem-solving.
- *Algorithmic and computational thinking skills*: Translate complex real-world problems into mathematical models for analysis and optimization.
- *Industry readiness*: Prepare students to build and implement robust systems for data-intensive fields including Scientific Computing, Machine Learning, Computational Finance, and Cryptography.
- *Research readiness*: Equip graduates with the expertise required to drive innovation in academic and industrial research.
- *Leadership*: Develop entrepreneurial skills and interdisciplinary competence for leadership roles in industry and academia.

3. Graduate Attributes

Graduates of the B.S. in Mathematics and Computing will:

- Demonstrate strong conceptual understanding and analytical proficiency in mathematics and computing.
- Utilize modern computational tools and technologies effectively across diverse multidisciplinary application domains using algorithmic thinking, mathematical rigor and logical reasoning.
- Integrate data-driven and mathematical approaches for informed analysis and decision-making.
- Ability to innovate and contribute towards next generation data driven technology.
- Communicate complex mathematical and computational ideas clearly and collaborate effectively within diverse and interdisciplinary teams.
- Exhibit leadership, professionalism, and initiative in academia, research, and industry.

4. Learning Outcomes

The students have:

- Rigorous training in mathematical reasoning across analysis, algebra, probability, linear algebra, and discrete structures.
- The ability to apply the mathematical concepts for designing efficient algorithms to solve application-oriented problems, analyze their complexity, and implement their computational solutions.
- The proficiency in building and analyzing mathematical models for data-driven systems using computational tools including quantum, physical and financial systems.
- The ability to utilize optimization techniques, statistical and machine learning approaches for data-driven decision-making.
- The ability to write technical reports, communicate mathematical and computational ideas effectively, and work ethically in teams.
- The ability to advance technological innovation and national growth by supporting Government-led-initiatives like Make in India, Startup India and Viksit Bharat 2047.

5. Eligibility Criteria & Admission Process

Admissions to B.S. Programs are made once a year (in July) through the all-India level Joint Entrance Examination (JEE) Advanced conducted by IITs. The procedures and other requirements for admission are specified in the JEE Information Brochure brought out every year.

6. Program Structure

S.N.	Course Type	Course Category	Credits	Minimum Total
1	Institute Core (I)	Engineering (IE)	16	39 (27.6%)
		Science (IS)	17	
		HSS (IH)	06	
2	Programme Core (P)	Programme Compulsory (PC)	66	90 (63.8%)
		Programme Electives (PE)	18	
		(B.Tech.) Project (PP)	6	
3	Open (O)	Electives (OE)	12	12 (8.5%)
			Total (Graded)	141 (100%)
4	Essential Audit (Non-taught)	Humanities (NH)	3	12
		Engineering (NE)	3	
		Industry-Academia Summer Internship outside IIT Jodhpur (two summers, minimum 45 days)	4	
		Design Credit	2	
		Total for award of Degree (Graded + Non-Graded)		153

6.1 Semester-wise Distribution of Credits

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	I Semester (ME/BB)/ II Semester (CS/EE)						**II Semester (ME/BB)/ I Semester (CS/EE)**				
IE	**Engineering Mechanics**	**3-0-0**	**3**	**-**	**3**	**IE**	**Intro. to Machine Learning**	**3-0-2**	**5**	**-**	**4**
IE	**Introduction to Computer Science**	**3-0-2**	**4**	**-**	**4**	**IS**	**Chemistry**	**3-0-0**	**3**	**-**	**3**
IS	**Physics**	**3-0-0**	**3**	**-**	**3**	**IS**	**Introduction to Bioengineering**	**2-0-2**	**4**	**-**	**3**
IH	**HSS I**	**3-0-0**	**3**	**-**	**3**	**IS**	**Chemistry Lab**	**0-0-2**	**2**	**-**	**1**
IS	**Physics Lab**	**0-0-2**	**2**	**-**	**1**						
IS	**Mathematics I**	**2-1-0**	**3**	**-**	**3**	**IS**	**Mathematics II**	**2-1-0**	**3**	**-**	**3**
IE	**Engineering Drawing**	**0-0-2**	**2**	**-**	**1**	**IE**	**Engineering Workshop**	**0-0-2**	**1**	**-**	**1**
NE	**Engineering Design I**	**0-0-2**	**2**	**1**	**-**	**NE**	**Engineering Design II**	**0-0-2**	**2**	**1**	**-**
NH	**Communication Skill I**	**0-0-2**	**2**	**1**	**-**	**IH**	**HSS II**	**3-0-0**	**3**		**3**

NH	Social Connect and responsibilities I	0-0-1	1	0.5	-	NE	Intro. To Profession	0-0-2	2	1	
NH	Performing Arts I /Sports I	0-0-1	1	0.5	-						-
Total				3	18	Total				2	18
	III Semester						IV Semester				
PC	Probability, Statistics and Stochastic Processes	3-1-0	4	-	4	PC	Differential Equations	3-0-0	3	-	3
PC	Data Structures and Algorithms	3-0-0	3		3	PC	Principles of Mathematical Analysis	3-0-0	3		3
PC	Discrete Mathematics	3-1-0	4		4	PC	Computational Linear Algebra	3-0-0	3		3
PC	Scientific Computations	3-0-0	3		3	PC	Signals and Systems	3-1-0	4		4
PC	Scientific Computations Lab	0-0-2	2		1	PC	Statistical Computation & Simulation	3-0-0	3		3
PC	CSE Lab I	0-0-2	2	-	1	PC	Artificial Intelligence	3-0-0	3	-	3
IE	Environment and Sustainability	3-0-0	3		3	PC	Computation and Simulation Lab	0-0-4	4		2
Total					19	Total				-	21
						Summer Internship				2	
	V Semester						VI Semester				
PC	Design & Analysis of Algorithms	3-1-0	4		4	PC	Applied Multivariate Analysis	3-0-0	3		3
PC	Optimization for ML	3-0-0	3		3	PC	Number Theory and Cryptography	3-0-0	3		3
PC	Theory of Computation	3-0-0	3		3	PC	Quantum Information	3-0-0	3		3
PC	Deep Learning	3-0-0	3		3	PC	Modern Algebra	3-0-0	3		3
PC	Optimization & DL Lab	0-0-4	4		2	PC	Applied Mathematics Lab	0-0-4	4		2
PE	Program Elective	3-0-0	3	-	3	PE	Program Elective	3-0-0	3		3
NH	Professional Ethics I	1-0-0	1	1	-						
Total				1	18	Total					17
						Summer Internship				2	

	VII Semester						VIII Semester				
PP	BS Project 1	0-0-3	-		3	PP	BS Project 2	0-0-3	-		3
PE	Program Electives	3-0-0	3	-	3	PE	Program Electives	3-0-0	3	-	3
PE	Program Electives	3-0-0	3	-	3	PE	Program Electives	3-0-0	3	-	3
OE	Open Electives	3-0-0	3		3	OE	Open Electives	3-0-0	3		3
OE	Open Electives	3-0-0	3		3	OE	Open Electives	3-0-0	3		3
Total					15	Total				-	15
Total of graded and Non-Graded Credit									6	141	
Non-Graded Design Credits									6	-	
Grand Total									12	153	

6.2 Detailed description of lab courses

S. No.	Lab Name	Lab components
1	Scientific Computations Lab	Scientific Computations Lab
2	CSE Lab I	Data Structures & Algorithms Lab
3	Computation and Simulation Lab	(i) Computational Linear Algebra Lab (ii) Statistical Computation & Simulation Lab
4	Optimization & DL Lab	(i) Optimization for ML (AIDE Lab III-3) (ii) Deep Learning Lab (AIDE Lab III-2)
5	Applied Mathematics Lab	(i) Applied Multivariate Analysis Lab (ii) Modern Algebra Lab

6.3 List of program elective courses

#	Mathematics	Computing	Statistics & Quantum Computing
1	Introduction to Game Theory (MAL4XXX)	Control Systems (EEL3XXX)	Introduction to Financial Engineering (MAL4XXX)
2	Graph Theory (CS4XXX)	Distributed systems (CS4XXX)	Quantum Computing (QCL4020)
3	Functional Analysis (MAL6XXX)	Introduction to Robotics (RM4XXX)	Multi-objective Optimization (MAL7XXX)
4	Partial Differential Equations (MAL6XXX)	Mathematical Fluid Dynamics (MAL6XXX) (new course)	Statistical Models & Regression (MAL7XXX)
5	Analytic Number Theory (MAL6XXX)	Advanced Algorithms (CS7XXX)	Stochastic Calculus (MAL7XXX)

6	Topology (MAL6XXX)	Computer Vision (CSLXXXX)	Time Series Analysis (MAL7XXX)
7	Numerical Methods for Partial Differential Equations (MAL7XXX)	Computer Graphics (CSLXXXX)	Quantum Inspired Optimization (QCL7XXX)
8		Digital Image Processing (EEL4XXX)	

6.4 Minor in Mathematics & Computing

Students aspiring to earn a *Minor* in *Mathematics and Computing* must successfully complete the courses listed below.

S. No.	Course Name	LTP	Credits
1	Statistical Computation & Simulation	3-0-0	3
2	Computational Linear Algebra	3-0-0	3
3	Computation and Simulation Lab	0-0-4	2
4	Applied Multivariate Analysis	3-0-0	3
5	Modern Algebra	3-0-0	3
6	Applied Mathematics Lab	0-0-4	2
7	Number Theory and Cryptography	3-0-0	3
Total			19

7. Targeted Students and their Career Prospects:

- **Targeted Students**

  The proposed B.S. program in Mathematics and Computing is designed for students with strong analytical aptitude and demonstrated interest in both mathematical reasoning and computational problem-solving. The program is particularly suited for students who:

  - Exhibit high proficiency in mathematics and logical thinking.
  - Aspire to integrate rigorous mathematical foundations with algorithmic and programming skills.
  - Are inclined toward abstraction, modeling, and theoretical problem-solving aligned with real-world applications and emerging technologies.
  - Seek preparation for research-oriented, data-intensive, and algorithm-centric careers.

- **Career Perspectives**

  The proposed program prepares graduates for careers in industry, research, higher studies, and emerging strategic sectors requiring strong mathematical and computational expertise.

  - *Industry Pathways*
    - Graduates will be well-suited for roles such as Data Scientist, Machine Learning Engineer,
    - AI Research Engineer, Algorithm Designer, Quantitative Analyst, Operations Research
    - Analyst, and Scientific Computing Professional. Opportunities span technology industries, fintech, R&D laboratories, consulting, defense, and other high-technology sectors.
    - The rigorous mathematical foundation ensures adaptability to rapidly evolving computational and AI-driven ecosystems.

  - *Higher Studies and Research*
    - The program provides strong preparation for M.Tech./MS/Ph.D. programs in Mathematics, Computer Science, Artificial Intelligence, Robotics, and Computational Sciences.
    - Graduates will be equipped for research in algorithms and complexity, distributed systems, optimization and control, statistical learning theory, computational geometry, and cryptography, enhancing their competitiveness for leading national and international graduate programs.

  - *Emerging and Strategic Domains*

    The program aligns with national and global priorities in Artificial Intelligence, autonomous systems, smart infrastructure, climate and sustainability analytics, computational biology, and cybersecurity, addressing the growing demand for mathematically trained computational professionals.

8. Faculty and Infrastructure Requirements

The minimum faculty (except probability and statistics) and infrastructure resources are already available with the Department.

9. Budgetary and Resource Requirements

1. **High-Performance Computing (HPC) Infrastructure:** To support the program core and elective courses such as Scientific Computing, Artificial Intelligence, Machine Learning, and Big Data, the program requires access to dedicated GPU-enabled servers within the institute's HPC ecosystem. These courses involve computationally intensive components, including simulations, large-scale data processing, and model training. Allocating dedicated server capacity for the B.S. M&C

program will help avoid queue delays arising from shared usage and ensure the smooth and timely conduct of lab sessions, projects, and evaluations.

2. **Department-Based Undergraduate Computing Lab:** We propose establishing a departmental UG computing lab equipped with essential programming and scientific software (C, C++, Python, MATLAB, etc.) to support B.S. M&C students beyond regular class hours. The lab will primarily serve students working on department-supervised projects, assignments, and independent computational work, while also providing a collaborative academic space within the department. This facility will complement the institute's central computing infrastructure and strengthen the program's focus on preparing students for research and industry-level computational challenges.

3. **Software Licenses and Subscriptions:** The program requires provision for licensed software, renewals, and specialized computational toolkits to support elective courses and project work. This includes maintaining and renewing existing institutional licenses (such as MATLAB and Mathematica) and enabling access to advanced libraries, cloud computing credits, data platforms, or domain-specific software as needed. While many core tools (such as Python, R, and C/C++) are open-source, budgetary flexibility will be important to ensure continued access to essential licensed and emerging technologies.

10. Industry & Research collaborations:

The B.S. in Mathematics and Computing programme will be strengthened through strategic collaborations with leading industries, financial institutions, deep-tech companies, and national research organizations to ensure continuous alignment with evolving technological demands and workforce requirements. Partnerships with IT and analytics firms, fintech and quantitative finance institutions, scientific and engineering industries, cybersecurity organizations, and emerging quantum and AI enterprises will provide opportunities for curriculum co-design, industry advisory support, internships, live capstone projects using real-world data, guest lectures, and practitioner-led electives. Collaborative engagements with national laboratories and premier academic institutions will promote undergraduate research in areas such as machine learning, optimization, cryptography, scientific computing, and quantum technologies. Industry-sponsored laboratories in Data Science, Computational Finance, Optimization, and Cybersecurity, along with innovation and incubation support, will further encourage entrepreneurship aligned with national initiatives such as Digital India, Make in India, Startup India, and Atmanirbhar Bharat. These interactions will ensure that graduates develop strong theoretical foundations alongside practical problem-solving skills, enhance employability and research readiness, and contribute effectively to high-impact roles in technology, finance, analytics, scientific computing, and emerging deep-technology sectors.

11. Expected Outcomes and impact:

The B.S. in Mathematics and Computing will generate significant long-term benefits for both students and society by building a strong pipeline of quantitatively trained and computationally proficient bunch of personnel who are capable of addressing complex, technology-driven challenges.

For students, the programme builds deep mathematical clarity along with practical computational skills that remains relevant across rapidly evolving domains such as artificial intelligence, data science, quantitative finance, cryptography, scientific computing, and advanced research. Graduates shall be a best fit for both software engineering and specialized research oriented roles in industry as well as academia. A strong mathematical background through this program will make it easier for them to transition into research oriented careers like PhD programs, specifically the roles involving cryptography or quantum computing. “Mathematical training” shall enhance their ability to solve complex, non-linear, and abstract problems that conventional “coding only” approaches might struggle with.

For society, the programme contributes to national capacity-building in critical areas including digital infrastructure, data-driven governance, financial systems, cybersecurity, and scientific innovation. These graduates can design reliable systems, analyze data responsibly, improve decision-making, and develop innovative solutions. In a world increasingly driven by algorithms and data, we need professionals who understand the mathematics behind the technology. With the rise of digital infrastructure, these graduates shall be vital in designing secure encryption methods and algorithms that protect financial, personal, and government data from cyber threats. Starting this programme will help create thoughtful, skilled, and responsible problem solvers who can contribute meaningfully to national development and global innovation.

Detailed Course Content of Programme Compulsory Courses

Title	**Mathematics I**	Number	MAL1XXX
Department	Mathematics	L-T-P [C]	2-1-0 [3]
Offered for	B.S., B.Tech. 1st Year (all branches)	Type	Institute Core
Prerequisite	None		

Objectives

1. Train the undergraduate students towards basic understanding of Mathematics.
2. Provide students with sufficient knowledge in calculus, which can be used by the students in their respective fields.
3. Enable students to develop a working knowledge of central ideas of single and multivariable calculus.

Learning Outcomes

1. Impart knowledge of basic concepts like Limit, Continuity, Differentiability, Maxima and Minima, Reimann Integration and their applications.
2. Gain strong understanding of Integration in higher dimension and Vector Calculus.
3. Improve their ability to think critically, to analyze a problem and solve it using a wide array of tools.

Contents

Single variable Calculus **[6 Lectures]*:***
Review of Sequences and series of real numbers, Review of Limit, Continuity and differentiability of functions, Rolle's theorem, Lagrange Mean value theorem and Taylor's theorem, Fundamental theorem of calculus, Application to length, area, volume, surface area of revolution.

Multivariable Calculus **[20 Lectures]:**
[7 Lectures] Functions of Several Variables, Sequences, Limit, and continuity of functions of several variables, Partial derivatives, Chain rule, Gradient, Directional Derivative and differentiability, Taylors formula for functions from R^2 to R, Tangent planes and normal, Maxima, minima, saddle points, Lagrange multipliers, Total derivative of Functions from R^n to R^m, Jacobian matrix, sufficient condition for differentiability.
[5 Lectures] Double and triple integrals, Change of variables, Vector fields, Gradient, Curl and Divergence, Curves.
[8 Lectures] Line integrals and their applications, Green's theorem: proof and applications, Surfaces, Surface area, Surface integrals, Divergence theorem, Stokes' theorem: proof and applications.

Text Books

1. Thomas, G. B. & Finney, R. L., (1992), *Calculus and Analytic Geometry,* 9th Edition, Addison Wesley Publishing Company
2. Ghorpade, S. R. & Limaye, B. V., (2006), *A Course in Calculus and Real Analysis,* Springer Verlag
3. Ghorpade, S. R. & Limaye, B. V., (2009), *A Course in Multivarible Calculus and Analysis,* Springer Verlag

Reference Books

1. Thomas Jr., G.B., Weir, M.D. and Hass, J.R., (2015), Thomas' Calculus, Pearson Education.
2. Apostol, T.M., (2006), Calculus: Vol-1 and Vol-2, John Wiley & Sons.
3. Ayers, F. and Mendelson, E. (2012) Schaum's Outline of Calculus, McGraw Hill.

Online Course Material

1. Dutta, J. (IIT Kanpur) Calculus of One Real Variable, https://nptel.ac.in/courses/109104124/2
2. Gupta, S.K. (IIT Roorkee) Multivariable Calculus, https://nptel.ac.in/courses/111107108/

Title	**Mathematics II**	Number	MAL1XXX
Department	Mathematics	L-T-P [C]	2-1-0 [3]
Offered for	B.S., B.Tech. 1st Year (all branches)	Type	Institute Core
Prerequisite	None		

Objectives

The Instructor will:

1. Provide a fundamental understanding of linear algebra.
2. Provide a fundamental understanding of ordinary differential equations.
3. Discuss the applications of linear algebra and differential equations.

Learning Outcomes

The students are expected to have:

1. knowledge of vector spaces and the properties like linear dependence, basis, and dimension.
2. understanding of linear transformations and its relation with matrices.
3. ability to analyze and solve ordinary differential equations.
4. ideas to apply linear algebra and differential equations on real life problems.

Contents

Linear Algebra **(17 Lectures):**

[9 Lectures] Vector spaces (over the field of real and complex numbers), Subspaces, Linear dependence/ independence, Basis, Dimensions, Coordinate with respect to a basis, Complementary subspaces, Linear transformations, Range space and rank, Null space and nullity, Matrix representation of linear transformation, Change of basis and similarity, Rank-nullity theorem.

[8 Lectures] Inner product, Norm, Gram-Schmidt orthogonalization process, Orthonormal bases, Eigenvalues and eigenvectors, Characteristic polynomials, Cayley-Hamilton theorem, Properties of eigenvalues and eigenvectors, Diagonalization.

Ordinary Differential Equations **(9 Lectures):**

[**6 Lectures**] First Order Equations: Basic definitions and geometrical interpretation, Solution by separation of variables, Exact equations, Integrating factors, Solution of IVP, Existence and uniqueness, Picard's theorem for IVP, Nonhomogeneous equations, Solutions of Bernoulli's, Riccati and Clairaut's equations.

[**3 Lectures**] Second Order Equations: Solution of equations with constant coefficient (CF & PI method), Wronskian, Euler-Cauchy equations, Variation of parameters.

Textbooks

1. Friedberg S. H., Insel A. J. and Spence L. E. (2015), Linear Algebra, Pearson
2. Kumaresan, S., (2000) Linear Algebra - A Geometric Approach, PHI Learning.
3. Simmons, G.F., (2002), Differential Equations with applications and Historical Notes, Tata McGraw Hill.

Reference Books

1. Strang, G., (2006), Linear Algebra and its Applications, Thomson, Brooks/Cole.
2. Axler, S., (1997), Linear Algebra Done Right (Undergraduate Texts in Mathematics), Springer.
3. Kreyszig, E., (2010), Advanced Engineering Mathematics, John Wiley & Sons
4. Coddington, E.A., (1961), An introduction to ordinary differential equations, Prentice-Hall.

Online Course Material

1. Prof I. K. Rana (IIT Bombay), Basic Linear Algebra, https://nptel.ac.in/courses/111101115/
2. Prof. K. C. Sivakumar (IIT Madras), Linear Algebra, https://nptel.ac.in/courses/111106051
3. S. Manam (IIT Madras), Differential Equations for Engineers, https://nptel.ac.in/courses/111106100/1

Title	**Probability, Statistics & Stochastic Processes**	Number	MAL2XXX
Department	Mathematics	L-T-P [C]	3-1-0 [4]
Offered for	B.S., B.Tech.	Type	Program Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Demonstrate the basic principles of probability theory and random variables.
2. Demonstrate the use of various families of probability distributions to model various types of data.
3. Introduce the concept of stochastic process and its application.

Learning Outcomes

The students will have the ability to:

1. Understand the probability theory and develops the mathematical foundation of random variables.
2. Understand random sampling, theory of estimation and testing of hypotheses.
3. Understand the concept of stochastic processes and demonstrate the specific applications to Poisson and Gaussian processes.

Contents

Probability Spaces and Random Variables [14 lectures]: Axioms of probability, conditional probability, Independence, Bayes' rule, Random variable, Some common discrete and continuous distributions, Distribution of Functions of Random Variable, Two and higher dimensional distributions, Functions of random variables, Conditional distributions, Laws of large numbers, Central limit theorem.

Estimation and Test of Hypothesis [11 lectures]: Parameter Estimation, Maximum Likelihood Estimator, Confidence Interval, Hypothesis Testing.

Stochastic Processes [14 Lectures]: Definition of Stochastic process, Classification and properties of stochastic processes, Simple Markovian stochastic processes, Google Page Rank, Gaussian processes, Stationary processes, Discrete and continuous time Markov chains.

Textbook

1. Ross, S. M., (2010), A First Course in Probability, Pearson Prentice Hall
2. Ross, S. M., (2012), Introduction to Probability and Statistics for Engineers and Scientists, Elsevier
3. Castaneda, L.B., Arunachalam, V. and Dharmaraja, S., (2012), Introduction to Probability and Stochastic Processes with Applications, John Wiley & Sons.
4. Papoulis, A., (1991), Probability, Random Variables and Stochastic Processes, McGraw-Hill.

Reference Books

1. Rohatgi, V. K. & Saleh, A.K.M.E., (2011), An Introduction to Probability and Statistics, Wiley
2. Casella, G. and Berger, R. (2002). Statistical Inference, Cengage Learning
3. Ross M.S., (2008), Stochastic Processes, John Wiley and Sons.
4. Hoel, P.G., Port, S.C. & Stone, C.J. (1986) Introduction to Stochastic Processes, Houghton Mifflin Company

Self-Learning Material

1. G. Srinivasan, IIT Madras, Introduction to probability and Statistics, https://nptel.ac.in/courses/111/106/111106112/
2. S. Dharmaraja, IIT Delhi, Introduction to Probability Theory and Stochastic Processes, https://nptel.ac.in/courses/111/102/111102111/

Course Title	**Discrete Mathematics**	Number	MAL2XXX
Department	Mathematics	L-T-P (C)	3-1-0 (4)
Offered for	B.S., B.Tech.	Type	Core
Prerequisite	None		

Objectives

1. Familiarize students with the fundamental concepts and principles of discrete mathematics.
2. Foster logical and analytical thinking skills. Establish a foundation for abstract mathematical understanding.
3. Prepare for practical applications in computer science and related fields.

Learning Outcomes:

The student will be able to:

1. To understand and assess the rigor of mathematical model definitions and conclusions, while recognizing and addressing fallacious reasoning and statements.
2. Apply the properties of combinatorial structures and employ basic techniques in combinatorics and counting.

Contents:

Mathematical Logic and Set Theory [11 Lectures]: Propositional logic, duality, propositional equivalences, satisfiability, predicates and quantifiers, rules of inference. Sets, functions, relations, equivalence relations, countability, partial and totally ordered sets, basic properties of lattices.

Proof Techniques [8 lectures]: Introduction to proofs, Well-Ordering Principle, direct proofs, proofs by contradiction, mathematical induction and recursion.

Combinatorics [12 Lectures]: Permutations and combinations, partitions, basic counting principles, sum rule, product rule, Inclusion-Exclusion Principle, Pigeonhole Principle, binomial coefficients and identities, recurrence relations, generating functions.

Graph Theory [8 Lectures]: Definition and properties of graphs, representing graphs using matrices, Handshaking Theorem, degree sequence, Special types of graphs: cyclic graph, complete graph, bipartite graph, planar graphs. Introduction to trees, characterizations of trees, graph matchings, graph colorings, Kempe chain arguments.

Textbooks:

1. Rosen, K.H., Discrete Mathematics and its applications, Eighth Edition, McGraw Hill, 2021.
2. Bóna, M., A Walk Through Combinatorics: An Introduction to Enumeration and Graph Theory, World Scientific, 2002.

Reference Books:

1. Lovász, L., J. Pelikán and K. Vesztergombi, Discrete Mathematics: Elementary and Beyond, Undergraduate Texts in Mathematics, Springer, 2003.
2. van Lint, J.H. and R.M. Wilson, A Course in Combinatorics, 2nd edition, Cambridge University Press, 2001.

Online Course Material

1. Iyengar, S., Discrete Mathematics, NPTEL Course Material, Department of Computer Science and Engineering, IIT Ropar, https://nptel.ac.in/courses/106106183
2. T. Leighton and M. van Dijk, Mathematics for Computer Science, Department of Computer Science & Engineering, MIT https://ocw.mit.edu/courses/6-042j-mathematics-for-computer-science-fall-2010/video_galleries/video-lectures/

Title	Scientific Computations	Number	MAL2XXX
Department	Mathematics	L-T-P-D [C]	3-0-0 [3]
Offered for	B.Tech ME, MT, CIE, CE	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Demonstrate the validity and error in the numerical results
2. Explain numerical techniques for a variety of problems
3. Introduce how to solve numerically algebraic equations, linear systems of equations, approximation and ordinary differential equations.

Learning Outcomes

The students will have the ability to:

1. Develop an understanding of numerical error and applicability of a particular method.
2. Solve numerically algebraic equations, linear systems of equations and ordinary differential equations.
3. Carry out numerical interpolation, differentiation and integration.

Contents

Errors in computation [Lecture 2]: Numerical Algorithms and errors, source and types of errors, error propagation, floating point representation, rounding error and floating point arithmetic.

Roots of equation [Lecture 7]: Iterative methods, order of convergence, Iterative methods for roots of nonlinear system of equations.

Linear systems of equations [Lecture 7]: Direct and iterative methods (Jacobi, Gauss Seidel), rate of convergence of iterative methods, Condition number, Power and inverse power methods for eigenvalue problems, matrix factorization schemes (LU, QR, spectral, Schur, polar, SVD).

Interpolation [Lecture 7]: Lagrange, Newton divided difference formula, Newton's interpolations, errors in interpolation.

Differentiation and Integration [Lecture 8]: differentiation using interpolation formulas, Integration using interpolation, Newton-Cotes formulas, Gauss quadrature rules, and Method of Undetermined Coefficients.

Ordinary differential equations [Lecture 8]: Taylor series method, Euler's Method, Modified Euler's Methods, multistep methods, Runge-Kutta Methods, adaptive methods, BVP finite difference methods, introduction to parallel computation.

Textbook

1. S.S. Sastry, (2012) Introductory Methods of Numerical Analysis, PHI Learning.
2. E. Kreyszig, (2010) Advanced Engineering Mathematics, Wiley.
3. S.C. Chapra, (2008) Applied Numerical Methods with Matlab for Engineers and Scientists, McGraw-Hill.

Reference Books

1. R.L. Burden and J.D. Faires, (2011) Numerical Analysis, Cengage Learning.
2. J.H Mathews and K.D. Fink, (2006) Numerical Methods using Matlab, Pearson Education.
3. R. Süli and D.F. Mayers, (2003) Introduction to Numerical Analysis, Cambridge University Press.

Self-Learning Material

1. Lal, R. and Banerjee S., Numerical Analysis, NPTEL Course Material, Department of Mathematics, IIT Roorkee, https://nptel.ac.in/courses/111107062/
2. Usha, R., Numerical Analysis, NPTEL Course Material, Department of Mathematics, IIT Madras, https://nptel.ac.in/courses/111106101/

Title	Scientific Computations Lab	Number	MAL2XXX
Department	Mathematics	L-T-P[C]	0-0-2 [1]
Offered for	B.Tech ME, MT, CIE, CE	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Equip students with practical skills to apply numerical techniques to real-world data and interpret results effectively.
2. Ability to implement models to solve numerically algebraic equations, linear systems of equations, approximation and ordinary differential equations.

Learning Outcomes:

The student will be able to:

1. Apply numerical methods to solve algebraic, matrix, and eigenvalue problems.
2. Implement interpolation, differentiation, and integration techniques for numerical approximation.

Solve ordinary and boundary value differential equations using computational tools and numerical algorithms.

Contents

Exp. No.	Experiment
1	Study of numerical errors, floating-point representation, rounding errors, and error propagation.
2	Solution of nonlinear equations using iterative methods (Bisection, Regula-Falsi, Newton-Raphson, Secant).
3	Solution of systems of nonlinear equations and convergence analysis of iterative methods.
4	Solution of linear systems using direct methods (Gaussian Elimination, Gauss-Jordan, LU Decomposition).
5	Solution of linear systems using iterative methods (Jacobi and Gauss-Seidel).
6	Computation of eigenvalues and eigenvectors using Power and Inverse Power Methods.
7	Implementation of matrix factorization techniques (LU, QR, Schur, Spectral, Polar, and SVD).
8	Polynomial interpolation using Lagrange and Newton Divided Difference methods.
9	Numerical differentiation using finite difference and interpolation-based formulas.
10	Numerical integration using Newton-Cotes formulas (Trapezoidal and Simpson's Rules).
11	Numerical integration using Gaussian Quadrature and Method of Undetermined Coefficients.
12	Solution of ODEs using Numerical Methods.

Textbook

1. S.S. Sastry, (2012) Introductory Methods of Numerical Analysis, PHI Learning.
2. E. Kreyszig, (2010) Advanced Engineering Mathematics, Wiley.
3. S.C. Chapra, (2008) Applied Numerical Methods with Matlab for Engineers and Scientists, McGraw-Hill.

Reference Books

1. R.L. Burden and J.D. Faires, (2011) Numerical Analysis, Cengage Learning.
2. J.H Mathews and K.D. Fink, (2006) Numerical Methods using Matlab, Pearson Education.
3. R. Süli and D.F. Mayers, (2003) Introduction to Numerical Analysis, Cambridge University Press.

Self-Learning Material

1. Lal, R. and Banerjee S., Numerical Analysis, NPTEL Course Material, Department of Mathematics, IIT Roorkee, https://nptel.ac.in/courses/111107062/
2. Usha, R., Numerical Analysis, NPTEL Course Material, Department of Mathematics, IIT Madras, https://nptel.ac.in/courses/111106101/

Course Title	**Differential Equations**	Number	MAL2XXX
Department	Mathematics	L-T-P (C)	3-0-0 (3)
Offered for	B.S.	Type	Core
Prerequisite	None		

Objectives

1. Explain and interpret the fundamentals of the existence and uniqueness of the solution of ordinary differential equations.
2. Apply and analyse analytical methods to solve ODEs and PDEs arising from mathematical and physical models.
3. Classify different types of partial differential equations and evaluate their solutions using analytical techniques for their suitability in various applications.

Learning Outcomes:

1. Demonstrate understanding of the theory, origin, and classification of ODEs and PDEs.
2. Solve and analyse ordinary and partial differential equations using appropriate analytical techniques.
3. Evaluate and interpret solutions of differential equations in the context of real-world problems and boundary value conditions.

Contents

Introductory Concepts: Introduction to Differential Equations through physical models, Solutions of linear homogeneous ODEs with constant coefficients, Exact Equations and integrating factors. **[3 lectures]**

Solutions by series method: Power series method to solve second-order linear homogeneous ODEs, Euler-Cauchy equations, Legendre Equation, Properties of Legendre polynomials, Classification of Singular points, Solution around a regular singular point- Frobenius Method, Bessel Equation and properties of Bessel functions.
[10 lectures]

Boundary Value Problems: Green's function, Sturm comparison theorem, Eigenvalue problems, Sturm-Liouville Theory. **[3 lectures]**

System of Ordinary Differential Equations: Linear systems, Algebraic properties of solutions of linear systems, The eigenvalue-eigenvector method of finding solutions, Fundamental matrix solutions, autonomous systems, stability analysis. Existence and Uniqueness theorems. **[9 lectures**]

First Order PDEs: Introduction and Classification, Method of characteristics for quasi-linear and nonlinear equations, Cauchy's problem, Cauchy-Kowalewski's Theorem. **[7 lectures]**

Second Order PDEs: Second-Order Partial Differential Equations: Classification, normal forms and characteristics, Laplace equation, Heat equation and wave equation: basic theory and solution by method of separation of variables. **[7 lectures]**

Textbooks:

1. Simmons, G. F., "Differential Equations with Applications and Historical Notes", CRC Press, 3rd Ed.
2. Boyce, W. E., DiPrima, R. C., and Meade, D. B., "Elementary Differential Equations and Boundary Value Problems", Wiley, 12th Ed.
3. Rao, K.S., "Introduction to Partial Differential Equations", PHI Learning Pvt. Ltd., 2nd Ed.
4. Strauss, W. A., "Partial Differential Equations: An Introduction", Wiley, 2nd Ed.

Reference Books:

1. Goode, S. W., and Annin, S. A., "Differential Equations and Linear Algebra", Pearson, 4th Ed.
2. Coddington, E. A., "An introduction to ordinary differential equations", Dover Publications Inc.

3. S. L. Ross, Differential Equations, 3rd Edn., Wiley India, 1984.
4. E. Zauderer, Partial Differential Equations of Applied Mathematics, 2nd Edition, John Wiley and Sons, 1989.

Online Course Material

1. NPTEL, Ordinary Differential Equations and Applications, IISc Bangalore, Nandakumaran, A. K., Datti, P. S., George, Raju K. https://nptel.ac.in/courses/111108081

2. NPTEL, Ordinary Differential Equations, IIT Kanpur, Raghavendra, V. https://nptel.ac.in/courses/111104031

3. NPTEL, First Course on Partial Differential Equations - I, IISc Bangalore, Nandakumaran, A. K., Datti, P. S., George, Raju K. https://nptel.ac.in/courses/111108144

Title	**Principles of Mathematical Analysis**	Number	MAL2XXX
Department	Mathematics	L-T-P [C]	3-0-0 [3]
Offered for	B.S.	Type	Core
Prerequisite	None		

Objectives:

1. To train students in the area of analysis.
2. Introduce abstract notions of convergence, compactness, and completeness in metric spaces.
3. Develop the theory of complex analytic functions and their consequences.

Learning Outcomes:

1. Metric Spaces, Completeness and Baire's Theorem.
2. Function Spaces, Pointwise and Uniform Convergence, Weierstrass M-Test.
3. Complex Differentiability, Cauchy-Riemann Equations, and Cauchy Integral Formula.

Content:

Metric Spaces [13 Lectures]: Definition and Examples, Open and Closed Sets, Continuous and Uniformly Continuous Functions, Convergence, Completeness, Banach Fixed Point Theorem, Connectedness, Compactness, Baire's Theorem and its Applications, Heine-Borel Theorem *(Theorems without proofs)*.

Sequences and Series of Functions [10 Lectures]: Sequences and Series of Functions, Pointwise and Uniform Convergence with Consequences (Continuity, Differentiability and Integrability), Cauchy Criterion for Uniform Convergence, Weierstrass M-Test, Power Series and Radius of Convergence.

Complex Analysis [16 Lectures]: Complex Functions, Continuity and Differentiation, Cauchy-Riemann Equations, Analytic Functions, Contour Integrals, Cauchy-Goursat Theorem, Cauchy Integral Formula, Identity Principle, Morera's Theorem, Liouville's Theorem, Fundamental Theorem of Algebra, Singularities, Laurent Series, Residue, Maximum Modulus Principle.

Textbooks:

1) S. Kumaresan, Topology of Metric Spaces, 2nd Ed., Narosa Publishing House, 2014.
2) R. G. Bartle and D. R. Sherbert, Introduction to Real Analysis, 4th Ed., John Wiley & Sons, 2015.
3) J. W. Brown and R. V. Churchill, Complex Variables and Applications, 9th Ed., McGraw-Hill, 2014.

Reference Books:

1. W. Rudin, Principles of Mathematical Analysis, 3rd Ed., McGraw-Hill, 1976.
2. T. M. Apostol, Mathematical Analysis, Narosa Publishing House, 2nd Ed., Addison-Wesley, 1974.
3. J. B. Conway, Functions of One Complex Variable, Vol. I, 2nd Ed., Springer-Verlag, 1978.
4. L. V. Ahlfors, Complex Analysis: An Introduction to the Theory of Analytic Functions of One Complex Variable, 3rd Ed., McGraw-Hill, 1979.

Online Course Material:

1. S. H. Kulkarni, Real Analysis, NPTEL Course Material, Department of Mathematics, IIT Madras, https://nptel.ac.in/courses/111106053/
2. P. A. S. Sree Krishna, Complex Analysis, NPTEL Course Material, Department of Mathematics, IIT Guwahati, https://nptel.ac.in/courses/111103070/

Course Title	**Computational Linear Algebra**	Number	MAL2XXX
Department	Mathematics	L-T-P (C)	3-0-0 (3)
Offered for	B.S.	Type	Core
Prerequisite	Calculus, Basic Linear Algebra / Matrix Theory, Programming		

Objectives

1. Study numerical methods for linear algebra in data and computational sciences.
2. Develop understanding of matrix factorizations and their implementations.
3. Enable application of linear algebra techniques to real-world data problems.
4. Build proficiency in efficient Python/MATLAB-based numerical computing.
5. Highlight the impact of computational cost and hardware on algorithm design.

Learning Outcomes

Students will be able to:

1. Analyze floating-point behavior, conditioning, and numerical stability.
2. Solve linear systems and least-squares problems with matrix factorizations
3. Perform PCA, ICA, and low-rank approximations using spectral methods.
4. Implement and interpret iterative solvers in optimization and ML contexts.
5. Solve large-scale linear algebra problems efficiently on modern hardware.

Contents

Numerical Foundations: Floating-point arithmetic, Sources of numerical error, Vector and matrix norms, Condition number, CPU architecture and computational cost. **[5 lectures]**

Linear Systems and Matrix Factorizations: Gaussian elimination, Jacobi and Gauss-Seidel methods,LU factorization, Cholesky factorization, Applications of Cholesky, Polar decomposition, Sparse matrices, Ridge regression as a regularized linear system, Error bounds for linear systems **[8 lectures]**

Orthogonalization based factorizations: Gram-Schmidt orthogonalization, Householder reflectors, Givens rotations, Comparison of Gram-Schmidt, Householder, and Givens QR, Least squares via QR, Weighted and regularized least squares, Optimization models **[7 lectures]**

Eigenvalue Problems and PCA: Eigenvalues and eigenvectors, Power method, inverse iteration, and shifted iteration, QR algorithm for eigenvalues, Principal Component Analysis and its applications, Independent Component Analysis. **[6 lectures]**

SVD, Lanczos Method, and Low-Rank Approximation: SVD, Lanczos algorithm, Truncated SVD and best low-rank approximation, SVD applications, Latent Semantic Analysis (LSA). **[6 lectures]**

Iterative Solvers and ML Optimization: Successive over-relaxation (SOR), Krylov subspace methods, Conjugate Gradient (CG) method, GMRES. **[7 lectures]**

Textbooks:

1) Datta, Biswa Nath, Numerical linear algebra and applications, PHI + SIAM, 2nd Ed, 2013.
2) Demmel, James W., Applied Numerical Linear Algebra, SIAM, 1997.

Reference Books:

1. Strang, G. Linear Algebra and Learning from Data Wellesley, Cambridge 2019
2. Trefethen, Lloyd N. and Bau III, David, Numerical linear algebra, SIAM, 1997.
3. Goodfellow, I., Bengio, Y. & Courville, A. Deep Learning, MIT Press, 2016.
4. Golub, Gene H., Matrix computations, The Johns Hopkins University Press, 2013

Online Course Material

1. MIT 18.065 – Matrix Methods in Data Analysis & ML (G. Strang) ocw.mit.edu/18-065S18. https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/

2. fast.ai – Computational Linear Algebra for Coders (R. Thomas) github.com/fastai/numerical- linear-algebra. https://www.fast.ai/posts/2017-07-17-num-lin-alg.html

Course Title	**Statistical Computation and Simulation**	Number	MAL3XXX
Department	Mathematics	L-T-P (C)	3-0-0 (3)
Offered for	B.S.	Type	Core
Prerequisite	Basic knowledge of probability & statistics		

Objectives

- Develop logical and computational thinking for tackling large-scale and complex statistical problems.
- Learn to use different simulation methods to study statistical properties and solve real-world problems.

Learning Outcomes:

The student will be able to:

- Implement techniques to generate data from common probability distributions.
- Apply methods like gradient ascent, MM and EM algorithms, stochastic gradient descent , Monte Carlo simulation and bootstrap techniques to solve complex statistical problems.
- Write programs in tools such as R programming language or Python to perform simulations.
- Integrate computational and statistical knowledge to address applied problems in data science, engineering, and business

Contents

Module-I: Pseudorandom number generation, sampling discrete and continuous random variables: inverse transform method, accept-reject algorithm, Box-Muller method, ratio-of-uniforms method, sampling from multivariate distributions, Simple and weighted importance sampling (**13 Lectures**)

Module-II: Gradient ascent, coordinate ascent; linear regression, ridge regression, lasso, logistic regression (**8 Lectures**)

Module-III: MM and EM algorithms, Gaussian mixture model, cross-validation and bootstrapping (**7 Lectures)**

Module-IV: Stochastic gradient descent, simulated annealing: examples and challenges, Bayesian models and Metropolis-Hastings algorithm, MCEM (11 **Lectures**)

Lab Components: Demonstration and implementation of algorithms corresponding to all methodologies discussed in Modules I-IV through R/Python

Textbooks:

1. Ross, S. M. (2012). Simulation. Academic Press
2. Robert, C. P., and Casella, G. (2004). Monte Carlo Statistical Methods. Springer.

Reference Books:

1. Efron, B., Tibshirani, R. J. (1994). An Introduction to the Bootstrap. CRC press.
2. Boyd, S., Boyd, S. P., Vandenberghe, L. (2004). Convex Optimization. Cambridge University Press.
3. Hastie, T., Tibshirani R. and Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference and Prediction. Springer.
4. Fishman, G.S. (1996). Monte Carlo: Concepts, Algorithms, and Applications. Springer.

Course Title	**Computation and Simulation Lab**	Number	MAL2XXX
Department	Mathematics	L-T-P (C)	0-0-2 (2)
Offered for	B.S.	Type	Core
Prerequisite	Statistical computation & Simulation, Linear Algebra		

Objectives

1. Study the implementation of numerical methods for linear algebra in data and computational sciences.
2. Develop understanding of matrix factorizations and their implementations.
3. Enable application of linear algebra techniques to real-world data problems.
4. Learn to use different simulation methods to study statistical properties and solve real-world problems.

Learning Outcomes

Students will be able to:

1. Solve linear systems and least-squares problems with matrix factorizations
2. Perform PCA, ICA, and low-rank approximations using spectral methods.
3. Implement and interpret iterative solvers in optimization and ML contexts.
4. Implement different statistical simulation and computational techniques to generate and analysis data from various probability distributions.

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Contents

(i) Computational Linear Algebra Lab

Lab	Lab Session Topic
1	Python/NumPy/SciPy setup; matrix operations, norms, solving Ax = b; condition number experiments
2	Floating-point arithmetic: cancellation, overflow, underflow; ill-conditioning effects on regression accuracy
3	LU factorization and pivoting from scratch; solving OLS normal equations; comparison with np.linalg.solve
4	Cholesky factorization; ridge regression implementation and regularization path; polar decomposition via SVD
5	Householder QR from scratch; orthogonality verification; numerical stability vs Gram-Schmidt
6	Givens rotations: implementation, selective zeroing, comparison with Householder; sparse matrix demo
7	Least squares regression: polynomial and multi-feature fitting; optimization model view; comparison with sklearn
8	Weighted least squares and ridge regression; bias-variance tradeoff on noisy datasets; CPU timing experiments
9	Eigendecomposition of covariance matrices; PCA from scratch on MNIST or Iris
10	ICA with FastICA: comparison of PCA and ICA on mixed-signal data
11	SVD; Lanczos algorithm; image compression using truncated SVD; reconstruction error vs rank
12	Jacobi/Gauss-Seidel solvers; convergence study; Conjugate Gradient on sparse SPD systems; GMRES demo

(ii) Statistical Computation and Simulation Lab

Lab	Lab Session Topic
1	Pseudorandom number generation
2	Sampling discrete and continuous random variables

3	Simple and weighted importance sampling
4	Gradient ascent, coordinate ascent
5	Regression
6	MM and EM algorithms
7	Gaussian mixture model, cross-validation and bootstrapping
8	Stochastic gradient descent, simulated annealing
9	Bayesian models and Metropolis-Hastings algorithm
10	MCEM

Textbooks:

1. Datta, Biswa Nath, Numerical linear algebra and applications, PHI + SIAM, 2nd Ed, 2013.
2. Demmel, James W., Applied Numerical Linear Algebra, SIAM, 1997.
3. Robert, C. P., and Casella, G. (2004). Monte Carlo Statistical Methods. Springer.

Reference Books:

1. Strang, G. Linear Algebra and Learning from Data Wellesley, Cambridge 2019
2. Trefethen, Lloyd N. and Bau III, David, Numerical linear algebra, SIAM, 1997.
3. Hastie, T., Tibshirani R. and Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference and Prediction. Springer.
4. Fishman, G.S. (1996). Monte Carlo: Concepts, Algorithms, and Applications. Springer

Online Course Material

1. MIT 18.065 – Matrix Methods in Data Analysis & ML (G. Strang) ocw.mit.edu/18-065S18. https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/

2. fast.ai – Computational Linear Algebra for Coders (R. Thomas) github.com/fastai/numerical- linear-algebra. https://www.fast.ai/posts/2017-07-17-num-lin-alg.html

Title	**Signals and Systems**	Number	EEL2XXX
Department	Electrical Engineering	L-T-P-D [C]	3-1-0 [4]
Offered for	B. Tech	Type	IE
Prerequisite			

Objectives

The Instructor will:

The Instructor would:

1. Familiarize students with the fundamentals of continuous-time and discrete-time linear systems, and their response analysis
2. Provide understanding frequency-domain transforms and their real-world applications
3. Discuss the applications from the area of engineering and physics

Learning Outcomes

The students would be able to:

1. Compute the response and system functions of LTI systems in various domains
2. Solve basic real-world interdisciplinary problems involving time-domain or frequency analysis of signals

Contents:

Signals and Systems (Continuous and Discrete Time) [10 lectures]: Description and Classification of signals, signal operations in time domain [4L],

Sampling, Description and Classification of LTI System, Properties of LTI system [4L],

Impulse response, Convolution and Correlation[2L].

Frequency Domain Representation of Signals [12 lectures]: Fourier series representation of periodic signals, convergence of Fourier Series & Gibbs phenomenon [3L],

Fourier Transform, Fourier transform of periodic signals, properties of Fourier transform [3L],

Laplace Transform, Z-Transform, region of convergence & properties [3L],

Rational System functions, Inverse Laplace and Z-Transform of Rational Functions and applications [3L]

LTI systems [12 lectures] : Difference and differential equation representation, block diagram representation [4L],

Transfer function and convolution, poles and zeros, step response [3L], Natural Response of 1st and 2nd Order Systems, Sinusoidal steady-state and Frequency Response [5L]

Applications: [8 lectures]: Applications from the area of condition monitoring, Predictive Operations and Maintenance, Process Control [4L],

ECG Signal Filtering and Denoising, Seismic Signal Processing, and few examples related to 2D,3D, and speech signal processing [4L]

Textbook

1. Oppenheim, A. & Wilsky, A. S., Signals and Systems, 2nd Edition, Prentice-Hall

Reference Books

1. Lathi, B. P., (2009), Principles of Linear Systems and Signals, 2nd Edition, Oxford University Press
2. Haykin, S. & Veen, B. V., (2008), Signals and Systems, 2nd Edition, Wiley
3. Dutta Roy, S. C., Chandra, S., (2018), Circuits, Systems and Signal Processing: A Tutorials Approach, 1st Edition, Springer

Self-Learning Material:

1. Dutta Roy, S. C., Indian Institute of Technology Delhi, Signals and Systems,https://www.youtube.com/playlist?list=PLC6210462711083C4
2. Venkatesan, K. S., Indian Institute of Technology Kanpur, Signals and Systems, NPTEL, https://nptel.ac.in/courses/117/104/117104074/

Title	**Artificial Intelligence (300)**	Number	CSL3XXX
Department	CSE, AI&DS	L-T-P-D [C]	3-0-0 [3]
Offered for	B.Tech (CSE, AI&DS)	Type	Compulsory
Prerequisite	Data Structures and Algorithms	Antirequisite	Artificial Intelligence (700) - CSL7XXX

Objectives:
To provide the foundations for AI problem solving techniques and knowledge representation formalisms.

Learning Outcomes:
The students are expected to have the ability to:
1. Identify and formulate appropriate AI methods for solving a problem.
2. Implement AI algorithms
3. Compare different AI algorithms in terms of design issues, computational complexity, and assumptions

Contents:

Introduction: Uninformed search strategies, Greedy best-first search, And-Or search, Uniform cost search, A* search, Memory-bounded heuristic search (5 Lectures)
Local Search Techniques: Beam Search, Hill Climbing Search, Genetic Search techniques (2 lectures)
Constraint Satisfaction Problems: Backtracking search for CSPs, Local search for CSPs (4 Lectures)
Adversarial Search: Optimal Decision in Games, The minimax algorithm, Alpha-Beta pruning, Expectimax search (5 Lectures)
Knowledge and Reasoning: Propositional Logic, Reasoning Patterns in propositional logic; First order logic: syntax, semantics, Inference in First order logic, unification and lifting, backward chaining, resolution (7 Lectures)
Representation: Information extraction, representation techniques, foundations of Ontology (4 Lectures)
Planning: Situation Calculus, Deductive planning, STRIPES, sub-goal, Partial order planner (4 Lectures)
Bayesian Network and causality: Probabilistic models, directed and undirected models, inferencing, reasoning, causality (6 lectures)
Reinforcement Learning: MDP, Policy, Q-value (5 Lectures)

Textbook
S. RUSSEL, P. NORVIG (2020), Artificial Intelligence: A Modern Approach, Pearson, 4th Edition

Reference Book :
1. E. RICH, K. KNIGHT, S. B. NAIR (2017), Artificial Intelligence, McGraw Hill Education, 3rd Edition.
2. J. PEARL (2009), Causality: Models, Reasoning and Inference, Cambridge University Press, 2nd Edition.
3. D. KOLLER, N. FRIEDMAN (2009), Probabilistic Graphical Models: Principles and Techniques, MIT Press

Course Title	**Applied Multivariate Analysis**	Number	MAL3XXX
Department	Mathematics	L-T-P (C)	3-0-0 (3)
Offered for	B.S.	Type	Core
Prerequisite	Basic knowledge of probability & statistics		

Objectives

- Develop understanding of multivariate data and methods for analyzing relationships among multiple variables.
- Equip students with practical skills to apply different multivariate techniques to real-world data and interpret results effectively.

Learning Outcomes:

The student will be able to:

- Understand and summarize multivariate data using statistical measures, graphical techniques, and matrix algebra.
- Analyze relationships among variables through covariance, correlation, and multivariate distributions.
- Apply key multivariate techniques such as PCA, factor analysis, classification, and clustering methods.
- Perform multivariate inference and interpret results using hypothesis testing methods (e.g., Hotelling's T^2, MANOVA) and statistical software.

Contents

Matrix Algebra and Random Vector: Some Basics of Matrix and Vector Algebra, Random Vectors and Matrices, Mean Vectors and Covariance Matrices, CDF and PDF of random vectors, Moments, Characteristics functions, Multinomial, Dirichlet. **(4 Lectures)**

Multivariate Normal Distribution: Normal data matrix (NDM): characterization and properties, Linear Transformation of NDMs, Wishart Distribution, The Hotelling's T^2 Distribution, Maximum likelihood estimation, Confidence Regions, Comparisons of several multivariate means, MANOVA (**10 Lectures)**

Discrimination Analysis: Fisher's LDA and QDA, Probabilities of misclassification **(5 Lectures)**

Principal Component Analysis: Principal components, Sampling properties of principal components, Principal component projections (**6 Lectures)**

Factor Analysis: The factor model, Principal factor analysis, Maximum likelihood factor analysis, Goodness of fit, rotation of factors, factor scores **(7 Lectures)**

Canonical correlation analysis: Population and sample canonical correlation vectors, variables and coefficients and their properties. **(7 Lectures)**

Textbooks:

1. Anderson, T. W. (2003). An Introduction to Multivariate Statistical Analysis. Wiley.
2. Johnson, R. A., and Wichern, D. W. (2019). Applied Multivariate Statistical Analysis. Pearson

Reference Books:

1. Hastie, T., Tibshirani R. and Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference and Prediction. Springer.
2. Muirhead, R.J. (2009). Aspects of Multivariate Statistical Theory. Wiley.
3. Mardia, K.V., Bibby, J.M., Kent, J.T. (1979). Multivariate Analysis, Academic Press.

Online Course Material

Applied Multivariate Analysis, Prof. Amit Mitra and Prof. Sharmishtha Mitra, IIT Kanpur.
https://www.youtube.com/watch?v=Ig2qF5BC8Fg&list=PL3DFCC23FCE3C7EFB

Course Title	**Number Theory and Cryptography**	Number	MAL3XXX
Department	Mathematics	L-T-P (C)	3-0-0 (3)
Offered for	B.S.	Type	Core
Prerequisite	None		

Objectives:

1. Develop foundational understanding of integers and divisibility.
2. Enhance logical reasoning and proof-writing skills.
3. Introduce applications of number theory in modern mathematics.
4. Discuss the fundamental ideas of symmetric and asymmetric key cryptography.

Learning outcomes: After successful completion of this course, students will be able to:

1. Apply concepts of divisibility, primes, modular arithmetic, and classical results such as Fermat's, Euler's, and Wilson's theorems.
2. Analyze and solve linear congruences and systems of congruences, and problems in modular arithmetic.
3. Understand the basic cryptographic algorithms and different cryptanalytic techniques.

Contents:

Divisibility and factorization [10 Lectures]: Well ordering principle, Principle of mathematical induction, Divisibility of integers, Euclid's algorithm, Prime numbers and their basic properties. Complete and reduced residue systems, Linear Diophantine equations.

Congruences and modular arithmetic [11 Lectures]: Linear congruences and their solutions, Chinese remainder theorem and its applications, Quadratic congruences, Classical results including Fermat's theorem, Euler's and Wilson's theorems, Quadratic residues, Legendre and Jacobi symbols, Primality testing algorithms.

Classical cryptography, Block and stream ciphers [10 Lectures]: Introduction to classical cryptography and their cryptanalysis, Perfect secrecy, Entropy, Iterated block ciphers, Substitution permutation networks, Feistel network, Basic concept of linear and differential attacks, Mode of operations, Stream ciphers.

Public key cryptography and Hash functions [08 Lectures]: RSA cryptosystem, Discrete logarithm problem, Diffie-Hellman key exchange, ElGamal cryptosystem, Signature scheme, Hash functions, Security of hash functions, Iterated hash functions

Textbooks:

1. Burton D. M. (2012), Elementary Number Theory. McGraw Hill Education.
2. Stinson D. R., Paterson M. B. (2019), Cryptography: Theory and Practice, CRC Press, 2019.

Reference books:

1. Niven I., Zuckerman H. S., Montgomery H. L. (1991), An Introduction to the Theory of Numbers, Wiley.
2. Paar C., Pelzl J. (2024), Understanding Cryptography, Springer.
3. Koblitz N. (1994), A Course in Number Theory and Cryptography, Springer.

Online course material:

1. Saikia A., Number Theory, https://nptel.ac.in/courses/111103020/.
2. Paar C., Introduction to Cryptography, https://www.youtube.com/@introductiontocryptography4223/featured

Course Title	**Modern Algebra**	Number	MAL3XXX
Department	Mathematics	L-T-P (C)	3-0-0 (3)
Offered for	B.S.	Type	Core
Prerequisite	None		

Objectives

1. To introduce fundamental concepts of groups, and rings with some applications.
2. To give sufficient knowledge of abstract algebra, required to do advanced courses in pure and applied mathematics.

Learning Outcomes:

The student will be able to:

1. precisely define terms like group, subgroup, normal subgroup, quotient group, ring, and ideal.
2. identify and provide examples of various algebraic structures, such as cyclic groups, Abelian Groups, Dihedral Groups, Symmetric and Alternating Groups, General Linear Groups, and rings.
3. apply the theory to solve a variety of problems, including finding the order of elements in groups, determining the structure of finite abelian groups, and determining whether a given group is simple.
4. apply the concept of group actions in counting techniques and coding theory.

Contents

Group Theory: Definition of groups, Examples and basic properties of groups, Subgroups, Cyclic groups, Abelian Groups, Dihedral Groups, Quaternion Group, Permutation Groups, Conjugacy Classes and Class Equation, Lagrange's Theorem, Normal Subgroups, Quotient groups, Group Homomorphism, Kernel and Image, Isomorphism theorems, Automorphism groups, Direct Products, Statement of Fundamental Theorem of finite abelian groups. **[20 lectures]**

Group action and Sylow Theorems: Group actions, Examples of group action, Sylow Theorems and their applications, overview of finite simple groups, Symmetry and counting, Applications in coding theory. **[10 lectures]**

Ring Theory: Definition of Rings, Examples and basic properties of rings, Subrings and Characteristics of a ring, Ideals, integral domain, Definition of field and Principal ideal domain, Quotient Rings, Ring Homomorphism, Isomorphism Theorems, Prime and maximal ideals, polynomial rings. **[9 lectures]**

Textbooks:

1. J. A. Gallian, Contemporary Abstract Algebra, 7th Edition, Brooks-Cole/Cengage Learning, 2010.
2. P. B. Bhattacharya, S. K. Jain, and S. R. Nagpaul, Basic Abstract Algebra, 2nd Edition, Cambridge University Press, 1994.
3. J. B. Fraleigh, A First Course in Abstract Algebra, 7th Edition, Addison-Wesley, 1982.

Reference Books:

1. D. S. Dummit and R. M. Foote, Abstract Algebra, 3rd Edition, John Wiley and Sons, 2003.
2. M. Artin, Algebra, Prentice Hall, 1991.
3. I. N. Herstein, Topics in Algebra, 2nd Edition, John Wiley and Sons, 1975.

Online Course Material

1. NPTEL, Introduction to Abstract Group Theory, Prof. Krishna Hanumanthu, Chennai Mathematical Institute. https://nptel.ac.in/courses/111106113

2. NPTEL, Introduction to Rings and Fields, Prof. Krishna Hanumanthu, Chennai Mathematical Institute. https://onlinecourses.nptel.ac.in/noc21_ma06/preview

Course Title	**Applied Mathematics Lab**	Number	MAL3XXX
Department	Mathematics	L-T-P (C)	0-0-4 (2)
Offered for	B.S.	Type	Core
Prerequisite	None		

Objectives

1. Equip students with practical skills to apply different multivariate techniques to real-world data and interpret results effectively.
2. To introduce students to computational methods in Modern Algebra using GAP for studying algebraic structures and their properties.
3. Strengthen analytical and logical reasoning skills by connecting theoretical algebra with computation and visualization using GAP.

Learning Outcomes:

The student will be able to:

1. Apply key multivariate techniques such as PCA, factor analysis, classification, and correlation analysis.
2. Perform computations involving groups, permutations, cosets, homomorphisms, and rings using GAP.
3. Apply concepts of Modern Algebra to puzzles and coding theory problems using computational techniques.

Contents

(i) Applied Multivariate Analysis Lab

S. No.	**Topic for lab session**
1	Random Vectors and Matrices
2	Normal data matrix
3	Wishart Distribution, The Hotelling's T^2 Distribution, Maximum likelihood estimation
4	Comparisons of several multivariate means, MANOVA
5	Fisher's LDA and QDA, Probabilities of misclassification
6	PCA
7	The factor model, Principal factor analysis, Maximum likelihood factor analysis
8	Goodness of fit, rotation of factors, factor scores
9	Canonical Correlation Analysis: Population and sample canonical correlation vectors
10	Canonical Correlation Analysis: Variables and coefficients and their properties

(ii) Modern Algebra Lab

S. No.	**Topic for lab session**
1	Why GAP? Introduction to GAP, and GAP group library

2	Permutations, Conjugates, and Commutator subgroups
3	Generators and Relations, Isomorphisms and Cosets
4	Groups of small order and their Subgroup lattices
5	Solution of Rubik's Cube using GAP
6	Group homomorphism, Automorphism, and Automorphism groups of Graphs
7	Group actions, Sylow Theorems, and Classification of Simple groups of small orders
8	Orbit-Stabilizer Theorem, and Counting techniques
9	The group of Sliding Block Puzzle
10	Rings and Coding Theory using GAP

Textbooks:

1. Anderson, T. W. (2003). An Introduction to Multivariate Statistical Analysis. Wiley.
2. Rosebrock S. (2024). Visual Group Theory, A computer-oriented geometric introduction, Springer.

Reference Books:

1. Hastie, T., Tibshirani R. and Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference and Prediction. Springer.
2. Muirhead, R.J. (2009). Aspects of Multivariate Statistical Theory. Wiley.
3. Mardia, K.V., Bibby, J.M., Kent, J.T. (1979). Multivariate Analysis, Academic Press.
4. Piterman K. I., Vendramin, L. (2023). Computer Algebra with GAP, Oberwolfach Preprints.
5. Holt D. F., Eick B., O'Brien E. A. (2020), Handbook of Computational Group Theory, CRC Press.
6. Grabmeier J., Kaltofen E., Weispfenning V. (2012) Computer Algebra Handbook: Foundations, Applications, Systems. Springer.

Online Course Material

1. Abstract Algebra in GAP, Alexander Hulpke. https://www.math.colostate.edu/~hulpke/CGT/howtogap.pdf

2. Groups : Motion, symmetry and puzzles, Prof. Amit Kulshrestha, IISER Mohali, https://onlinecourses.nptel.ac.in/e-learning/preview/noc20_ma28

Detailed Course Content of Programme Elective Courses

Course Title	**Mathematical Fluid Dynamics**	Number	MAL6XXX
Department	Mathematics	L-T-P (C)	3-0-0 (3)
Offered for	B.S., B.Tech., M.Sc.	Type	Elective
Prerequisite	Differential Equations and Vector Calculus		

Objectives

1. To introduce students to the fundamental laws and equations of fluid dynamics.
2. To analyze several specific fluid flow systems and their respective applications.
3. To provide students a rigorous mathematical understanding of fluid behavior and instabilities.

Learning Outcomes:

1. Students will gain an understanding of the mathematical principles underlying fluid dynamics.
2. Students will be able to learn fundamental concepts on various fluid flow systems and solve different physical/engineering problems.
3. Students will be acquainted with linear/non-linear stability analysis of fluid flow.

Contents

Module-1[10-lectures]: Definitions and classification of fluid flows, Kinematics-Lagrangian and Eulerian description, Equation of continuity, Boundary surface, Streamlines, Path lines and streak lines, Equations of Motion-Lagrange's and Euler's equations of motion, Bernoulli's theorem, Equation of motion by flux method, Equations referred to moving axes, Impulsive actions, Stream function.

Module-2[10 lectures]: Velocity potential, Irrotational and rotational motions, Vortex lines. Irrotational motion in two-dimensions, Complex velocity potential, Sources-sinks-doublets, Milne-Thomson circle theorem, Two-dimensional irrotational motion produced by motion of circular, co-axial and elliptic cylinders, Kinetic energy of liquid, Theorem of Blasius, Motion of a sphere through a liquid at rest at infinity, Liquid streaming past a fixed sphere, Stokes stream function.

Module-3[10 lectures]: Stress components in a real fluid, Relations between rectangular components of stress, Connection between stresses and gradients of velocity, Viscous incompressible flows, Navier-Stoke's equations, Dimensional analysis and similarity. Plane Poiseuille and Couette flows, Flows at small Reynolds numbers, Flow through uniform cross-sections. Unsteady flows.

Module-4[9 lectures]: Perturbation Analysis, Method of normal modes, inviscid theory, Kelvin- Helmholtz instability, Rayleigh's equation, stability of viscous flows, Orr-Sommerfeld equation, Squire's theorem, Energy equation, Stability of special flows.

Textbooks

1. P. K. Kundu and I. M. Cohen, Fluid Mechanics (3rd edition), Elsevier Science & Technology, 2002.
2. W. O. Criminale, T. L. Jackson and R. D. Joslin, Theory and Computation in Hydrodynamic Stability, Cambridge University Press (Second Edition), 2018.

Reference Books

1. F. Chorlton, Textbook of Fluid Dynamics, CBS Publishers, 1998.
2. F. M. White, Fluid Mechanics, McGraw-Hill, New York, 8th Ed., 2015.
3. A. J. Chorin and J.E. Marsden, A Mathematical Introduction to Fluid Mechanics, Springer-Verlag, 1999.
4. P. J. Schmid and D. S. Henningson, Stability and Transition in Shear Flows, Springer-Verlag New York (Applied Mathematical Sciences), 2001.

Online Course Material

1. S. Chakrabarty, Introduction to Fluid Mechanics, NPTEL Course Material, Mechanical Engineering, IIT Kharagpur,https://nptel.ac.in/courses/112105269
2. G M Homsy et al, Multimedia Fluid Mechanics Online, Cambridge University Press https://www.cambridge.org/core/homsy/
3. V. Shankar, Fluid Mechanics, NPTEL Course Material, Chemical Engineering, IIT Kanpur, https://nptel.ac.in/courses/103104044

Course Title	**Analytic Number Theory**	Number	MAL6XXX
Department	Mathematics	L-T-P (C)	3-0-0 (3)
Offered for	B.S., M.Sc.	Type	Elective
Prerequisite	Complex Analysis		

Objectives:

1. Develop a strong foundation in arithmetic functions and prime distribution.
2. Apply analytic techniques, including the theories of Dirichlet series and L-functions, to study number-theoretic problems.
3. Introduce modular forms and explicate their role in modern analytic number theory.

Learning Outcomes: After successful completion of the course students will be able to:

1. Apply analytical methods to study partial sums and the distribution of primes.
2. Define fundamental objects such as the Gamma function, Theta functions, the Riemann Zeta function, Dirichlet L-functions, and study their properties.
3. Demonstrate knowledge of basics of modular forms and their applications in number theory.

Contents:

Arithmetic Functions and Distribution of Primes (12 Lectures): Introduction and definitions, Multiplicative functions, Dirichlet convolution, Möbius inversion formula, Big-O notation, Average orders and partial sums of arithmetic functions, Euler-Maclaurin summation formula, Chebyshev functions, Prime Number Theorem and its equivalent forms, and Ikehara-Weiner Tauberian theorem (without proof).

Special Functions (5 Lectures): Gamma function, Bernoulli polynomials, Mellin and Inverse Mellin transform, Bessel functions (definitions + basic properties only), Theta Functions and Perron's formula.

Dirichlet Series and Euler Products (14 Lectures): Character Groups, Dirichlet Characters, Dirichlet Series, Primes in Arithmetic Progressions, Multiplication of Dirichlet Series, Euler Products, Theory of Riemann Zeta Functions (Location of Poles and Zeros, Functional Equation, Special Values, analytic continuation via Poisson or Euler-Meclaurin summation formula).

Modular Forms (8 Lectures): Modular group, Basic definitions, Eisenstein series, q-expansions, and Space of modular forms.

Textbooks:

1. Apostol, T. M., Introduction to Analytic Number Theory, Springer-Verlag, 1976.

2. Apostol, T. M. Modular Functions and Dirichlet Series in Number Theory (2nd ed.). Graduate Texts in Mathematics, Vol. 41. Springer-Verlag New York, 1990.

Reference Books: Iwaniec, H., & Kowalski, E. Analytic Number Theory. American Mathematical Society Colloquium Publications, Vol. 53. Providence, RI: American Mathematical Society, 2004.

Course Title	Finite Element Methods: Theory and Computation	Number	MAL6XXX
Department	Mathematics	L-T-P (C)	3-0-0 (3)
Offered for	B.S., M.Sc., M.Tech.	Type	Elective
Prerequisite	Linear algebra, Partial Differential equations, Scientific Computing		

Objectives

1. To introduce the mathematical foundations of the FEM
2. To develop the ability to construct and implement FEM algorithms for 1D and 2D BVPs
3. To understand error analysis, convergence, and stability in numerical solutions of PDEs.
4. To expose students to various FEM techniques, their real-world applications, and tools.

Learning Outcomes:

The student will be able to

1. Formulate boundary value problems in weak form and identify appropriate function spaces.
2. Develop and implement FEM algorithms for solving 1D and 2PDEs
3. Analyze error, convergence behavior, and conditioning of FEM solutions.
4. Apply FEM to practical problems and use modern computational tools for simulation.

Contents

Motivation & Mathematical Foundations (9 Lectures): Real-world problems, Limitations of finite difference, irregular domains, FEM philosophy, strong vs Weak Formulation, motivation for weak form, Function Spaces, Sobolev spaces, Energy norms, Variational Formulation, Bilinear forms, Linear functionals, Existence and Uniqueness of solution of variational formulation, Lax-Milgram theorem

1D FEM Construction (12 Lectures): Domain Discretization, Mesh in 1D, Nodes and elements, Basis Functions, Piecewise linear basis, Hat functions, Galerkin Method, Projection idea, FEM as Galerkin, Assembly of System, Stiffness matrix, Load vector, Imposing Boundary Conditions, Dirichlet vs Neumann implementation, Full 1D FEM Algorithm, Error & Convergence, Approximation error intuition, Numerical Integration, Gaussian quadrature, hands on 1D FEM

2D FEM Fundamentals (12 Lectures): 2D Domains and Meshes, Triangular meshes, Mesh generation tools, Basis Functions in 2D, Linear elements (triangles), Mapping to Reference Element, Affine transformations, Element Integrals, Jacobian, Numerical quadrature, Assembly in 2D, Global matrix construction, Boundary Conditions in 2D, Full 2D FEM Algorithm, hands on 2D FEM

Error Analysis & Theory (6 Lectures): Interpolation Error, Approximation properties, Céa's Lemma, Convergence Rates, Mesh size h, Conditioning & Stability, Ill-conditioning, Sparse matrices

The course will also integrate the FEM software ecosystem—FEniCS, deal.II, and COMSOL Multiphysics etc—to demonstrate real-world applications and computational workflows.

Text books:

1. S. Ganesan and L. Tobiska, *Finite elements: Theory and Algorithms*, Cambridge University Press, 2017
2. C. Johnson, *Numerical solutions of partial differential equations by the finite element method*, Cambridge University Press, 1987

Reference Books:

1. S. C. Brenner and R. Scott, *The Mathematical Theory of Finite Element Methods*, Springer, 3rd ed., 2008
2. M. S. Gockenbach, *Understanding and Implementing the Finite Element Method*, SIAM, 2006

Online Course Material

1. Prof. Krishna Garikipati and Prof. Gregory Teichert, University of Michigan, https://open.umich.edu/find/open-educational-resources/engineering/introduction-finite-element-methods#materials
2. Prof. Biswanath Banerjee and Prof. Amit Shaw, IIT Kharagpur, https://onlinecourses.nptel.ac.in/noc26_ce42/preview

Title	**Stochastic Calculus**	Number	MAL7XXX
Department	Mathematics	L-T-P [C]	3-0-0 [3]
Offered for	B.S., M.Sc., Ph.D.	Type	Elective
Prerequisite			

Objectives:

1. To develop an understanding of financial market
2. Given a discrete process, pricing of options, such as European, American, Asian and other exotic options
3. To develop an understanding of change of measure, Martingales and Brownian motion
4. Use of Black-Scholes model under various conditions

Learning Outcomes:

1. Understanding derivatives and securities
2. Pricing and valuation in discrete case
3. Binomial model and self-financing strategy
4. Black Scholes model and interest rate models

Contents :

[9 lectures]: General probability spaces, Gaussian and log-normal distribution, martingales and stopping times, Brownian motion and its properties, Ito's integral and martingale properties of Ito's integral.

[12 lectures]: Continuous time financial market models, Black-Scholes-Merton model, Black-Scholes PDE and formulas, risk-neutral valuation, option pricing, the Greeks, putcall parity, Risk-neutral valuation, risk-neutral measure, Girsanov's theorem for change of measure, martingale representation theorems, representation of Brownian martingales

[18 lectures]: Stochastic differential equations, existence and uniqueness of solutions, Feynman-Kac formula and its applications, models with stochastic volatility, pricing and hedging in incomplete markets, bond markets, term-structures of interest rates, bond pricing, Short rate models, martingale models for short rate, Vasicek, Cox-Ingersoll-Ross and Hull-White models, multifactor models; Forward rate models, Heath-Jarrow-Morton model.

Textbooks:

1. Shreve, S. E., Stochastic Calculus for Finance I; The Binomial Asset Pricing Model , Springer-Verlag, 2004
2. Shreve, S. E., Stochastic Calculus for Finance II; Continuous-Time Model, Springer, 2004. **Reference Books:**

1. Baxter, M. and Rennie, A., Financial Calculus, University Press, Cambridge 1996

Title	**Introduction to Game Theory**	Number	MAL7XXX
Department	Mathematics	L-T-P [C]	3-0-0 [3]
Offered for	B.S., B.Tech.	Type	Elective
Prerequisite			

Objectives:

1. To provide basic understanding of Game Theory and its applications.
2. To equip the students with sufficient knowledge in Game theory which can be used by the students in their respective fields.

Learning Outcomes:

1. Understanding of basic of Non-cooperative games and Nash Equilibria
2. Understanding of Dual Simplex method and concept of Duality with applications to game theory.
3. Understanding of Nash Model with security point, stable strategies and Bayesian Games.

Contents :

Game Trees, Choice Functions and Strategies, Choice Subtrees, Equilibrium N-tuples Strategies. (4 Lectures)

Normal Forms, Non-cooperative games, Nash Equilibrium and its computation, The von Neumann Minimax Theorem, Mixed strategies, Best Response Strategies. (8 Lectures)

Matrix Games and Linear Programming, Simplex Algorithm, Avoiding cycles and Achieving Feasibility, Dual-Simplex Algorithm, Duality Theorem. (9 Lectures)

2x2 Bimatrix Games, Nonlinear Programming Methods for Non-zero Sum Two-Person Games, Coalitions and Characteristic Functions, Imputations and their Dominance. (4 Lectures)

The Core of a game, Strategic Equivalence, Stable Sets of Imputations, Shapley Values, N-Person Non-Zero Sum Games with continuum of strategies - Duels, Auctions. (8 Lectures)

Nash Model with Security Point, Threats, Evolution, Stable Strategies, Population Games, Bayesian Games. (6 Lectures)

Textbooks:

1. M. MASCHLER, E. SOLAN, S. ZAMIR (2013), Game Theory, Cambridge University Press.
2. M.J. OSBORNE, A. RUBINSTEIN (1994), A Course in Game Theory, The MIT Press.

Reference Books:

1. P.D. STRAFFIN (Jr.) (1993), Game Theory and Strategy, The Mathematical Association of America.

Online course material:

1. M.O. JACKSON, K.L. BROWN, Y. SHOHAM, Game Theory, Coursera Course Material, https://www.coursera.org/learn/game-theory

Title	**Numerical Methods for Partial Differential Equations**	Number	MAL7XXX
Department	Mathematics	L-T-P [C]	3-0-0 [3]
Offered for	B.S., M.Tech., P.hd	Type	Elective
Prerequisite	Probability, Statistics and Random Processes		

Objectives

The Instructor will introduce

1. finite difference schemes for numerical solutions of partial differential equations
2. carry out numerical analysis of various finite difference schemes

Learning Outcomes

The students are expected to have the ability to:

1. apply finite difference schemes for numerical solution of partial differential equations
2. carry out error analysis and stability analysis of finite difference schemes

Contents

Fractal 1: Numerical methods for PDE (MAL8XX1)

[7 Lectures] Characterization of PDEs, Finite difference methods for elliptic, parabolic and hyperbolic problems.
[6 Lectures]Stability, consistency and convergence theory, dissipation and dispersion, error estimates.

Fractal 2: Numerical Analysis for PDE (MAL8XX2)

Parabolic equations [6 Lectures]: Fourier analysis of the error, Alternating direction implicit method
Elliptic equations [6 Lectures]: Stability analysis by the energy method, Error analysis with a maximum principle, asymptotic error estimates, convergence result
Hyperbolic equations[8 Lectures]: CFL condition, Von Neumann stability Analysis, upwind scheme, Lax-Wendroff scheme, leap-frog scheme, Hamiltonian systems and symplectic schemes
Numerical linear algebra[6 Lectures]: Classical iterative Methods for solving linear systems, large sparse linear systems, GMRES algorithm, preconditioned Conjugate Gradient method, multigrid method (overview), Newton's method and its variations for solving nonlinear systems.

Textbook

1. Quarteroni, A. and Valli, A. , Numerical Approximation of Partial Differential Equations, Springer, 1997
2. Morton, K. W. and Mayers, D. F. , Numerical solution of partial differential equations, CUP, 2005

Reference Books

1. Thomas, J. W. , Numerical partial differential equations: Finite difference methods, Springer, 1995 2. Trefethen, L. N. and David Bau III, Numerical Linear Algebra, SIAM, 1997

Online course Material

1. Nayak, A. K., Numerical methods: Finite Difference Approach, NPTEL Course, Department of Mathematics, IIT Roorkee, https://nptel.ac.in/courses/111107107/.

Title	**Time Series Analysis**	Number	MAL7XXX
Department	Mathematics	L-T-P [C]	3-0-0 [3]
Offered for	B.S., M.Tech.	Type	Elective
Prerequisite	Probability, Statistics and Random Processes		

Objectives

1. To provide working knowledge of time series techniques and forecasting methods
2. To provide with techniques and receipts for estimation and assessment of quality of econometric models with time series data

Learning Outcomes

1. To develop the skills needed to do empirical research in fields operating with time series data sets

Contents

Models for Time Series[7 Lectures] Time series data, trend, seasonality, cycles and residuals, strong and weak stationarity, autocorrelation function, linear processes, estimation of mean and covariance functions,Wold decomposition Theorem.

Models of stationary process [7 Lectures]ARMA (p, q) processes, ACF and PACF, Modeling using ARMA processes, estimation of parameters, testing model adequacy, Order estimation.

Univariate Forecasting Models [7 Lectures] Prediction in stationery processes, special reference to ARMA processes, Frequency domain analysis – spectral density and its estimation, transfer functions.

Multivariate Forecasting Models [8 Lectures]Single equation models, Vector AR and ARMA models, econometric models

Non-stationary Models [10 Lectures]Stationarity through differencing, ARIMA model, ARMAX, ARIMAX models and introduction to ARCH models.

Indicative Assignments: Analysis of real data (namely, NEON data, IMD data, etc.) using different time series models

Textbook

1. Blockwell, P. J. and Davis, R. A. (2017). Introduction to Time Series and Forecasting, 2nd Edition, Springer.

2. Chatfield, C. (2004) The Analysis of Time Series – An Introduction, Chapman and Hall / CRC, 4thed.

Reference Book:

1. Box, G.E.P., Jenkins, G. and Reinsel, G. (1994) Time Series Analysis-Forecasting and Control, 3rd ed., Pearson Education

Online material:

Mikusheva, Anna, Time Series Analysis, Department of Economics, Massachusetts Institute of Technology, MIT OpenCourseware Course Material, https://ocw.mit.edu/courses/economics/14-384-time-series-analysis-fall-2013/recitations/

Title	**Partial Differential Equations**	Number	MAL6XXX
Department	Mathematics	L-T-P [C]	3-0-0 [3]
Offered for	B.S., M.Sc.	Type	Compulsory
Prerequisite			

Objectives

1. Introduce the concepts of existence and uniqueness of solution of partial differential equations.
2. Develop analytical techniques to solve partial differential equations.
3. Understand the properties of the solution of partial differential equations.

Learning Outcomes

1. Understanding existence, uniqueness, and other properties of a solution of partial differential equations.
2. Solving partial differential equations with Fourier series method, Green's function method, Laplace transform method, and Fourier transform method.

Contents

Motivations and origins [3 Lectures]: PDEs of engineering and physics (Heat, wave and Laplace equations).

First order partial differential equations [9 Lectures]: Linear, quasi-linear and fully nonlinear equations, Lagrange method, the Cauchy problem, Cauchy method of characteristics, compatible systems, Charpit method.

Second order partial differential equations [11 Lectures]: Classification and Canonical forms, Adjoint operators, Riemann's method, solving linear partial differential equations with constant coefficients.

Elliptic, parabolic and hyperbolic equations[16 Lectures]: Variable separation method for Laplace/Poisson equation, heat conduction equation, wave equation, Burger equation, D'Alembert's solution, Maximum-Minimum principles, Duhamel's principle, solutions by Green's function method.

Text Books

1. Rao, K. S., Introduction to partial differential equations, 3rd Edition, PHI, 2011.
2. Logan, J. D., Applied partial differential equations, 3rd Ed., Springer, 2015.
3. Pinchover, Y. and Rubinstein, J., An introduction to partial differential equations, Cambridge University Press, 2005.

Reference Books

1. Sneddon, I. N., Elements of Partial Differential Equations, McGraw-Hill, New York,1986.
2. Prasad, P. and Ravindran, R., Partial differential equations, New Age International Publisher, 2005.
3. Mattheij, R. M. M., Rienstra, S. W., and ten Thije Boonkkamp, J. H. M., Partial Differential Equations: Modeling, Analysis, Computation, SIAM Publication, 2005.

Online Course Material

1. Agrawal, P. N. and Pandey D.N., Ordinary and partial differential equations and applications, NPTEL Course Material, Department of Mathematics, IIT Roorkee, https://nptel.ac.in/courses/111107111/
2. Sinha, R. K., Partial differential equations, NPTEL Course Material, Department of Mathematics, IIT Guwahati, https://nptel.ac.in/courses/111103021/

Title	**Multi-objective Optimization**	Number	MAL7XXX
Department	Mathematics	L-T-P [C]	3-0-0 [3]
Offered for	B.S., M.Tech.	Type	Elective
Prerequisite			

Objectives

1. Introduce multi-objective optimization.
2. Introduce evolutionary computations and using these to solve nonlinear optimization problems.
3. Application of multi-objective optimization.

Learning Outcomes

1. Understand the ideas of evolutionary computations and multi-objective optimization.
2. Understand the advantages of multi-objective optimization.
3. Solving single and multi-objective optimization problems using evolutionary computations.
4. Implementation of multi-objective optimization in some engineering problems.

Contents

Evolutionary optimization [10 lectures]: Introduction to evolutionary optimization, genetic algorithms, mathematical models of genetic algorithms. Evolutionary programming and evolution strategies, ant colony optimization, particle swarm optimization.

Multi-objective Optimization [7 lectures]: Partial ordering with respect to a pointed cone, non-dominated sorting, introduction to multi-objective optimization, solution of a multi-objective optimization problem, efficient, weak efficient, and proper efficient solution, ideal and nadir vector, applications of multi-objective optimization in engineering.

Classical methods for solving multi-objective optimization [6 lectures]: Weighted sum method, epsilon-constrained method, goal programming technique, weighted metric methods, lexicographic ordering method. Advantages and limitations of classical methods.

Multi-objective optimization using evolutionary algorithm [8 lectures]: Evolutionary algorithms for multi-objective optimization - binary and real coded genetic algorithm NSGA-II, NSGA-III, particle swarm optimization for multi-objective optimization, implementation of evolutionary algorithms.

Application of Multi-objective optimization in polynomial time solvable problems and NP hard problems [8 lectures]: Shortest path problems, spanning trees, assignment problems, Knapsack problems, branch and bound problems, and travelling salesman problems.

Textbooks:

1. Dan Simon, Evolutionary Optimization Algorithms, Wiley, 2013
2. K. Deb, Multiobjective optimization using evolutionary algorithms, Wiley, 2003
3. M. Ehrgott, Multicriteria optimization, Springer, 2005

Reference Books:

1. R. Fletcher, Practical Methods of Optimization, 2nd Edition, Wiley, 2000
2. K. Miettinen, Nonlinear Multiobjective Optimization, Kluer Academic Publications, 1999

Online Course Material

1. D. Sharma, Evolutionary Computation for Single and Multiobjective Optimization, Department of Mechanical Engineering, IIT Guwahati, NPTEL course materials, https://nptel.ac.in/courses/112/103/112103301/
2. K. Deb, Introduction to Optimization and its Scope in Practice, GIAN - MHRD, IIT Kharagpur video lectures, https://www.youtube.com/watch?v=4mZp_-SApks&list=PLBNAhOTc16q9CyTPoJzrklUQE3zPujwyf&t=2s

Title	**Topology**	Number	MAL6XXX
Department	Mathematics	L-T-P [C]	3-1-0 [4]
Offered for	B.S., M.Sc.	Type	Compulsory
Prerequisite			

Objectives

1. To train the students in the domain of Topology.
2. To give sufficient knowledge of the subject which can be used by students for further applications in their respective domains of interest.

Learning Outcomes

1. Topological spaces, order topology and product topology.
2. Separation axioms, countability axioms Urysohn's metrization theorem.
3. Function spaces, pointwise and uniform convergence, Stone-Weierstrass theorem.

Contents

Basic Topology [14 Lectures]: Topological spaces, basis and sub-basis, order topology, product topology, subspace topology, convergence, closed sets, continuous functions, quotient topology

Properties of Topological Spaces [16 Lectures]: Connected spaces, components, local connectedness, compact spaces, limit point compactness, local compactness, countability axioms, separation axioms, normal spaces, Tychonoff spaces, Urysohn's metrization theorem, Tietze extension theorem, Tychonoff theorem, Stone-Cech compactification

Special Topics [09 Lectures]: Function spaces, pointwise and uniform convergence, compact-open topology, uniform convergence on compact sets, Ascoli’s theorem, Stone-Weierstrass theorem.

Textbooks

1. Munkres, J., Topology, second edition, Pearson Publications, 2000
2. Willard, S., General Topology, Dover Publications, 2004

Reference Books

1. Pervin, W. J. and Boas, R. P., Foundations of general topology, Academic Press.
2. Simmons, G. F., Introduction to topology and modern analysis, Tata McGraw-Hill, 2004

Online Course Material

1. Veeramani, P., Topology, NPTEL Course Material, Department of Mathematics, IIT Madras, https://nptel.ac.in/courses/111106054/.

Title	**Functional Analysis**	Number	MAL6XXX
Department	Mathematics	L-T-P [C]	3-1-0 [4]
Offered for	B.S., M.Sc.	Type	Compulsory
Prerequisite			

Objectives

1. To introduce the area of Functional analysis.
2. To give sufficient knowledge of the subject which can be used by student for further applications in their respective domains of interest.

Learning Outcomes

1. Basic idea of normed linear spaces and operators on normed linear spaces.
2. Open Mapping theorem, Hahn-Banach Theorem, and their applications.

Contents

Normed Linear Spaces and Banach spaces [10 Lectures]: Normed linear space, Equivalent norms, Banach spaces, Linear operator, norm of a linear operator, boundedness and continuity, Quotient spaces.

Bounded linear operators on function spaces [15 Lectures]: l_p spaces, L_p spaces, uniformly bounded operator, principle of uniform boundedness, Open mapping theorem, Closed graph theorem, Hahn-Banach Theorem, dual of a normed linear space, separability and reflexivity of normed linear space, weak convergence, weak* convergence

Hilbert Spaces and their Applications [9 Lectures]: Inner product spaces, polarisation identity, Bessel's inequality, Hilbert space, Riesz representation theorem, Operators on Hilbert spaces, compact operators, projections and idempotents.

Introduction to spectral theory [5 Lectures]: Eigenvalues of linear operators, spectrum of bounded linear operators, spectral properties of bounded linear operators

Text Books

1. Limaye, B. V., *Functional Analysis*, Second Edition, New Age International, 1996
2. Kesavan, S. *Functional Analysis*, Second Edition, Springer, 2023

Reference Books

1. Conway, J. B., *A course in Functional Analysis*, Second Edition, Springer-Verlag, 1994
2. Kreyzig, E., *Introduction to Functional Analysis with Applications*, John Wiley & Sons, New York, 1978.
3. Vaidyanathan, P., *Functional Analysis*, Cambridge University Press, 2023.

Online Course Material

1. Srivastava, P. D., Functional Analysis, NPTEL Course Material, Department of Mathematics, IIT Kharagpur, https://nptel.ac.in/syllabus/111105037/